

Ahmed Abdullah

ML Engineer — Data Engineer

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EXPERIENCE

- **Indexy Inc.** Ontario, CA (Remote)
Software Engineer, AI Infrastructure *Feb 2026 – Present*
 - **Data Infrastructure Engineering:** Led 3-engineer team building core ML and data infrastructure; optimized a high-scale ClickHouse logging pipeline, cutting monthly infrastructure costs by 75%. Built a query fan-out data modeling pipeline on GCP with a 0% failure rate. Developed headless browser infrastructure reducing scraping failure rates to <1%. Contributed internal APIs for scalable scraping of major LLM platforms (ChatGPT, Claude, Grok, DeepSeek, Meta AI).
- **Progyny Inc.** New York, US (Remote)
Data Analytics Engineer, Healthcare *Jun 2025 – Feb 2026*
 - **ML and Data Platform Engineering:** Designed and scaled ETL and ML data pipelines using Apache Airflow, Marauder, and Dagster; integrated Salesforce, Snowflake, and unstructured document sources via Python and SQL. Reduced pipeline latency from 60+ seconds to ~3 seconds, a **95%** improvement, enabling production ML and personalization workflows at scale.
- **MacroMed (Pvt. Ltd.)** Lahore, PK
Machine Learning Engineer *Feb 2025 – Jun 2025*
 - **Production Conversational AI and Recommendation Systems:** Led 3-engineer team to design and deploy a production-grade conversational AI platform using the RAFT framework with Mistral Nemo for low-latency LLM inference. Built real-time recommendation pipelines on PostgreSQL achieving sub-2s end-to-end latency and ~0.5s retrieval; deployed full-stack on DigitalOcean with a Next.js frontend.
- **Multimedia Mining and Search Group, Johannes Kepler University** Linz, AT (Remote)
Research Engineer, Multimodal Learning *Aug 2024 – Feb 2025*
 - **Multimodal Representation Learning:** Developed deep learning models for face-voice association under missing modality conditions, improving accuracy by **12%** on the VoxCeleb benchmark using WavLM, ECAPA, and VGGVox. Redesigned training pipeline with reusable modules, reducing processing time by **10%** and experiment setup time by **70%**.

EDUCATION

- **National University of Computer and Emerging Sciences** Lahore, Pakistan
Bachelor of Science in Data Science; GPA 3.6 *Aug. 2022 – May. 2026*
 - **Honors:** Dean's List of Honor, Top of Batch
 - **Experience:** Teaching Assistant for OOP (CS1004) and COAL (EE2003)

PROJECTS

- **Rhazys – Explainable Tool for Medical Imaging:** Built an explainable AI platform for medical imaging in collaboration with King Edward Medical University, publishing 3 papers in MICCAI with SOTA results. Designed interpretable deep learning pipelines to support clinical decision-making, combining model performance with transparency for real-world medical use.
- **DermaScan3D – Skin Cancer Detection using CNNs:** Built a multimodal CNN-based skin cancer classification system using TensorFlow and ResNet architectures, ranking in the top 1% among 8000+ participants on Kaggle through optimized preprocessing and model tuning.

PROGRAMMING SKILLS

- **Languages:** Python, SQL, C++.
- **Technologies:** PyTorch, TensorFlow, Transformers, OpenCV, Scikit-learn, MONAI.
- **Tools & Platforms:** LangChain, Amazon Web Services, Google Cloud Platform, Apache Airflow, Snowflake, Salesforce, Odoo, Huspot, PostgreSQL, Docker, MLflow, ClickHouse, Playwrite, NextJS, ReactJS.